Chapter 4: Real-time Estimation of Spatially Correlated 1-D Drag Coefficient

4.1 Introduction

The hexapole actuation system is a 6-input 3-output system. Since the soft magnetic material is used in the fabrication of the hexapole system, it is subject to the hysteresis effect. With the derivation and implementation of Hall-sensor-based magnetic flux control and optimal flux control, the 6-input hexapole actuation system can be treated as a linear decoupled 3-axis force producer with a bandwidth close to 4000Hz, which is experimentally verified in chapter 2. A model-based position-dependent control law was derived and implemented in chapter 3 to achieve high-speed untethered manipulation of the magnetic microprobe in aqueous solutions, where the motion error is a zero-mean random error, dictated by random thermal forces and random measurement noise. We can achieve minimum variance control simultaneously.

The drag coefficient characterized by Stoke's law at low Reynold's number is not constant in aqueous solutions. As the probe moves close to a wall, the drag coefficient varies with its distance from the surface. The position-dependent drag coefficient is characterized by two principal directions, perpendicular and parallel to the wall [58,59]. In chapter 3, the experiment result is based on the update of drag coefficient using the theoretical value with known wall height. In this chapter, we derived a Kalman filter-based augmented estimator [56] that estimates the time-variant and/or spatially dependent input gain (drag coefficient) in the absence of the information of wall height. It enables us

to realize adaptive time variant control when propelling the probe in aqueous solution accounting for wall effects.

4.2 Wall Effect and Spatially Correlated Drag Coefficient

The drag coefficient is time-variant and/or spatially dependent due to the wall effect. As the probe approaches a wall, the drag coefficient varies with its distance from the surface. The position-dependent drag coefficient is characterized by two principal directions, perpendicular and parallel to the wall. The theoretical value is experimentally verified in Fig. 13. The control law (3.6) that we proposed in chapter 3.3 may cause instability without real-time update of the control gain (drag coefficient).

If we use the nominal drag coefficient in (3.6), let the nominal input gain $a_N = \Delta t/\gamma_N$ and $a[k] = \Delta t/\gamma[k]$, wherein γ_N is the nominal drag coefficient and $\gamma[k]$ is the time-varying coefficient, and define $e_a[k] = a[k] - a_N$, which is the parameter error in (3.6), the error dynamic for the case of measurement delay $d = 2$ is cast as follows:

$$\delta x[k+1] = \lambda_c \cdot \delta x[k] - \frac{e_a[k]}{a_N}(1-\lambda_c)\delta x[k-2] \tag{4.1}$$

where $\varepsilon[k]$ is ignored. Define c as the ratio between the two drag coefficients, i.e., $c = \gamma[k]/\gamma_N$. Jury's stability test [57] is performed to derive the stability range of c with respect to λ_c. The result is summarized as follows.

$$\begin{cases} \dfrac{2(1-\lambda_c)}{2-3\lambda_c+\sqrt{{\lambda_c}^2+4}} < c, & 0 \le \lambda_c < 1 \\ \dfrac{(1-\lambda_c)}{2} < c < \dfrac{(1-\lambda_c)}{-\lambda_c}, & -1 < \lambda_c < 0 \end{cases} \tag{4.2}$$

In the case of $\lambda_c = 0$, c must be greater than 0.5 determined by (4.2) to ensure stability. That is, if the probe is controlled and moved to a region where the medium has a lower drag coefficient, the motion of the probe has a chance to become unstable. A computer simulation experiment is performed to validate the analytical prediction of the stability range. In the experiment, the microbead is controlled to move from one region having $\gamma = 4.25 \times 10^{-8} (N \cdot s/m)$ to the other region having lower drag coefficient and then move back to the original position. The control law is designed according to the known drag coefficient and the designed value of λ is zero, i.e., $\lambda = 0$. The left panel of Fig. 27 shows the result of motion control for the case of $c = 0.49$, wherein the boundary of the two regions is depicted by the blue line. It is seen that the motion of the microbead becomes unstable after entering the second region but becomes stable again after returning to the first region. The right panel of Fig. 28 shows the result of the motion control for the case of $c = 0.51$. Although the motion remained stable after entering the second region, the variance of the tracking error increased significantly.

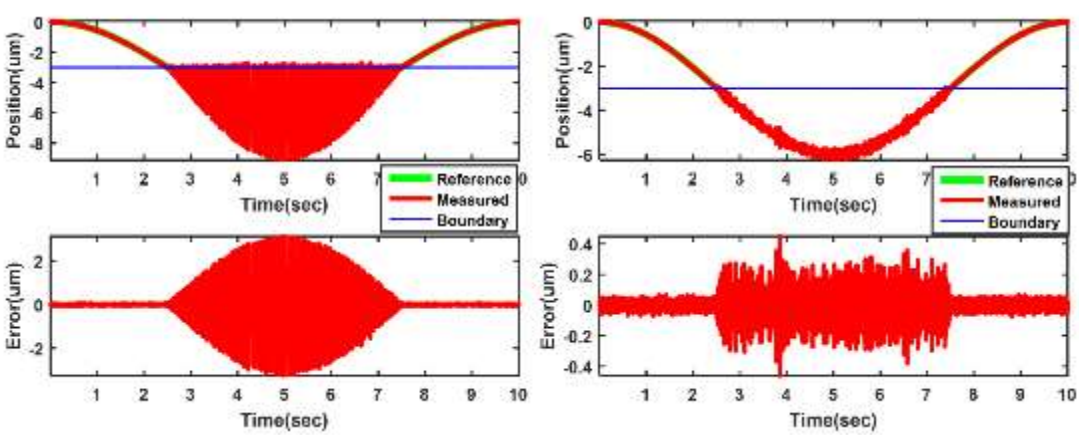

Fig. 28. The left panel shows the resulting motion of $c = 0.49$, and the right panel shows resulting motion of $c = 0.51$.

4.3 Real-time Estimation of Drag Coefficient and Control Law Adaptation for Untethered Manipulation

4.3.1 Kalman-filter-based Augmented Estimator

Considering the time-variant and/or spatially dependent drag coefficient, (3.3) can be rewritten as,

$$x[k+1] = x[k] + a_x[k](f_d[k] + f_T[k]) + x_D[k] \tag{4.3}$$

wherein $a_x[k] = \frac{\Delta t}{\gamma[k]}$, f_d is the desired force determined from the control law, f_T is the thermal force experienced by the probe, $x_D[k] = a[k]D[k]$ is motion due to the lumped of the modeling error term and the disturbance, and $D[k]$ is the lumped of the modeling error term and the disturbance. Let the control objective be $\delta x[k+1] = \lambda_c \cdot \delta x[k]$, where $\delta x[k] = x_d[k] - x[k]$, and $x_d[k]$ is the desired motion trajectory, comparing (4.3) with the desired motion, the force required to meet the control objective can be obtained,

$$f_d[k] = \frac{1}{\hat{a}_x[k]}\{x_d[k+1] - x_d[k] + (1-\lambda_c)\delta\hat{x}[k] - \hat{x}_D[k]\} \tag{4.4}$$

where $\hat{a}_x[k]$ is the estimated input gain, $\delta\hat{x}[k]$ is the estimated tracking error, and $\hat{x}_D[k]$ is the estimated motion due to the lumped of the modeling error term and the disturbance. Substituting (4.4) into (4.3) and comparing the resulting motion with the desired motion trajectory, the dynamics of the tracking error can be derived,

$$\delta x[k+1] = \lambda_c \cdot \delta x[k] + (1-\lambda_c)e_{\delta x}[k] - e_a[k]f_d[k] - e_{x_D}[k] - a_x[k]f_T[k] \tag{4.5}$$

where the estimation errors are $e_{\delta x}[k] = \delta x[k] - \delta\hat{x}[k]$, $e_a[k] = a_x[k] - \hat{a}_x[k]$, and $e_{x_D}[k] = x_D[k] - \hat{x}_D[k]$

The measured tracking error is defined as $\delta x_m[k] = \delta x[k-d] + n_x[k]$. It consists of two parts which are the deterministic term and the random term which is subject to the random measurement noise and thermal force. Our objective is to use (3.11) which is the relationship between the variance of the tracking error and the input gain to predict the input gain. First of all, we use a single-pole Infinite Impulse Response (IIR) filter to estimate the mean of the measured tracking error.

$$\overline{\delta x}_m[k+1] = a_{var} \cdot \delta x_m[k] + (1 - a_{var})\overline{\delta x}_m[k] \tag{4.6}$$

where $(1 - a_{var})$ is the pole location which determines the effective length of calculating the mean of the measured tracking error. The random term of the measured tracking error is defined as $\delta x_r[k] = \delta x_m[k] - \overline{\delta x}_m[k]$. We use the same IIR filter (4.6) to get the mean and the mean square of $\delta x_r[k]$. The definition of the variance is the mean square minus the squared mean, i.e.,

$$\sigma^2_{\delta x_r}[k] = \overline{x_r^2}[k] - \overline{x_r}^2[k] \tag{4.7}$$

The relationship between the tracking error variance and the input gain is described in (3.11). However, we are using the measured tracking error, so we need to add an additional measurement noise to (3.11).

$$\sigma^2_{\delta x_r}[k] = \left(2 + \frac{1}{1-\lambda_c^2}\right) 4k_b T a_x[k] + \left(\frac{2}{1+\lambda_c}\right)\sigma^2_{n_x} \tag{4.8}$$

Thus, the prediction of the input gain can be determined by rearranging (4.8)

$$a_x[k] = \frac{\left(\sigma^2_{\delta x_r}[k] - \left(\frac{2}{1+\lambda_c}\right)\sigma^2_{n_x}\right)}{\left(4k_b T\left(2 + \frac{1}{1-\lambda_c^2}\right)\right)} \tag{4.9}$$

This is one of the components of the output of our augmented estimator. The other output is the measured tracking error.

Kalman filter-based augmented estimator [56] is designed to estimate time-variant and/or spatially dependent drag coefficient in real-time for drag correction to render adaptive self-tuning control when propelling the probe in aqueous solutions accounting for wall effects. Based on the tracking error (4.5), the Kalman filter-based augmented estimator can be derived as,

Let $\delta x_1[k] = \delta x_m[k], \delta x_2[k] = \delta x_m[k+1], \delta x_3[k] = \delta x_m[k+2],\ x_D[k+1] = x_D[k] + \delta x_D[k],\ \delta x_D[k+1] = \delta x_D[k],\ a_x[k+1] = a_x[k] + \delta a_x[k],\ and\ \delta a_x[k+1] = \delta a_x[k]$

The prediction of the state is

$$\underbrace{\begin{bmatrix} \delta\hat{x}_1^f[k+1] \\ \delta\hat{x}_2^f[k+1] \\ \delta\hat{x}_3^f[k+1] \\ \hat{x}_D^f[k+1] \\ \delta\hat{x}_D^f[k+1] \\ \hat{a}_x^f[k+1] \\ \delta\hat{a}_x^f[k+1] \end{bmatrix}}_{\hat{\boldsymbol{X}}^f[k+1]} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & \lambda_c & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \underbrace{\begin{bmatrix} \delta\hat{x}_1[k] \\ \delta\hat{x}_2[k] \\ \delta\hat{x}_3[k] \\ \hat{x}_D[k] \\ \delta\hat{x}_D[k] \\ \hat{a}_x[k] \\ \delta\hat{a}_x[k] \end{bmatrix}}_{\hat{\boldsymbol{X}}[k]} \tag{4.10}$$

The output of the estimator is $\boldsymbol{y}[k] = \begin{bmatrix} \delta x_m[k] \\ a_x[k] \end{bmatrix}, \hat{\boldsymbol{y}}[k] = \begin{bmatrix} \delta\hat{x}_m[k] \\ \hat{a}_x[k] \end{bmatrix}$

The estimation error dynamic is

$$\underbrace{\begin{bmatrix} e_{\delta x_1}[k+1] \\ e_{\delta x_2}[k+1] \\ e_{\delta x_3}[k+1] \\ e_{x_D}[k+1] \\ e_{\delta x_D}[k+1] \\ e_{a_x}[k+1] \\ e_{\delta a_x}[k+1] \end{bmatrix}}_{\boldsymbol{E}[k+1]} = \underbrace{\begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & -f_d[k] & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}}_{\boldsymbol{F}[k]} \underbrace{\begin{bmatrix} e_{\delta x_1}[k] \\ e_{\delta x_2}[k] \\ e_{\delta x_3}[k] \\ e_{x_D}[k] \\ e_{\delta x_D}[k] \\ e_{a_x}[k] \\ e_{\delta a_x}[k] \end{bmatrix}}_{\boldsymbol{E}[k]} \tag{4.11}$$

The prediction of the error covariance is

$$\boldsymbol{P}^f[k] = \boldsymbol{F}[k]\boldsymbol{P}[k-1]\boldsymbol{F}^T[k] + \boldsymbol{Q}[k-1] \tag{4.12}$$

where $\boldsymbol{P}[k-1]$ is the error covariance matrix, and $\boldsymbol{Q}[k-1]$ is the variance of the process noise.

The estimator gain matrix is

$$\boldsymbol{K}[k] = \boldsymbol{P}^f[k]\boldsymbol{H}^T(\boldsymbol{H}\boldsymbol{P}^f[k]\boldsymbol{H}^T + \boldsymbol{R}_k)^{-1} \tag{4.13}$$

where $\boldsymbol{R}_k$ is the variance of the measurement noise, and $\boldsymbol{H} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$

Update the estimation through the estimator gain

$$\widehat{\boldsymbol{X}}[k+1] = \widehat{\boldsymbol{X}}^f[k+1] + \boldsymbol{K}[k](\boldsymbol{y}[k] - \widehat{\boldsymbol{y}}[k]) \tag{4.14}$$

Update the error covariance

$$\boldsymbol{P}[k] = \frac{1}{\lambda_f}[\boldsymbol{I} - \boldsymbol{K}[k]\boldsymbol{H}]\boldsymbol{P}^f[k] \tag{4.15}$$

where λ_f is the forgetting factor which gives exponentially less weight to older error samples.

4.3.2 Experimental Verification of the Drag Coefficient Estimation

The Kalman filter-based augmented estimator is designed to real-time estimate the drag coefficient (input gain) and realize adaptive time variant control. The adaptive time variant control is applied to solve the stability issue in Fig. 28. In the simulation experiment, the control law (4.4) is implemented. The result of the motion is shown in the left panel of Fig. 29. The real-time estimation of the drag coefficient is shown in the right panel of Fig. 29. The motion is stabilized in the second region and followed the reference trajectory. The estimated drag coefficient followed the sudden change of the real drag coefficient at the boundary.

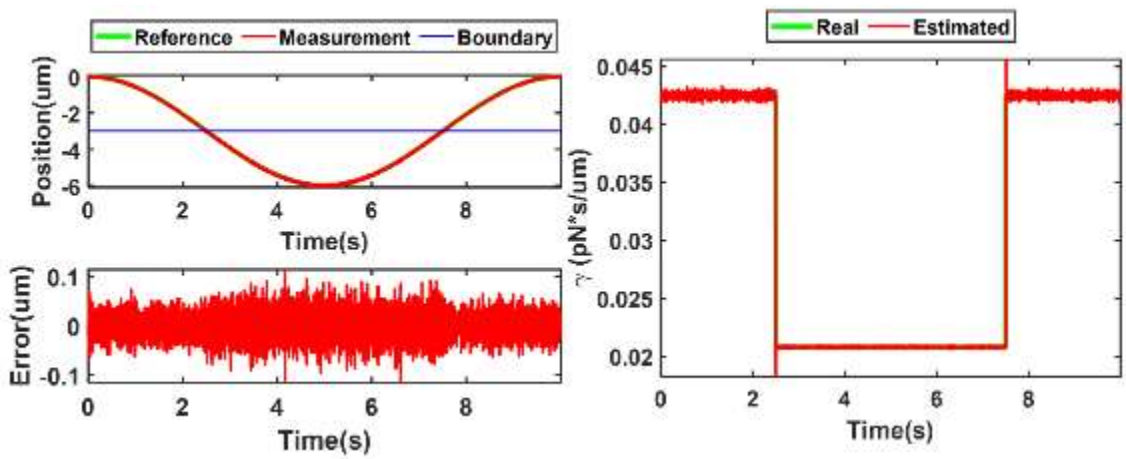

Fig. 29. The left panel shows the resulting motion with $\hat{a}_x[k]$, and the right panel shows the estimation of the drag coefficient.

The second experiment is performed to demonstrate the importance of the real-time drag coefficient estimation with respect to the tracking performance. The probe started from the initial position of $(0\ \mu m, 0\ \mu m, 0\ \mu m)$. The z motion of the probe is a 12.5-micrometer ramp to the negative direction of z, which is approaching the wall. The

motion of x and y are the same. They are 1 Hz sinusoidal waves with 9 micrometers amplitude. The total motion lasted for 10 seconds. The probe is stabilized and controlled by the control law (3.6). Instead of using the real drag coefficient, we use the nominal drag coefficient γ_N in (3.6). In the experiment, the wall is placed at $z = -15$ micrometers and the microbead is immersed in water. Since the ramp is 12.5 micrometers, the closest distance from the probe center to the wall is 2.5 micrometers. The radius of the probe is 2.25 micrometers. The left panel of Fig. 30 shows the result of motion without using real-time estimation. The right panel of Fig. 30 shows the corresponding tracking error. It is shown that the tracking error gradually increased as the probe approached the wall. The increase in the tracking error is due to the increase in the difference between the real drag coefficient and the nominal drag coefficient.

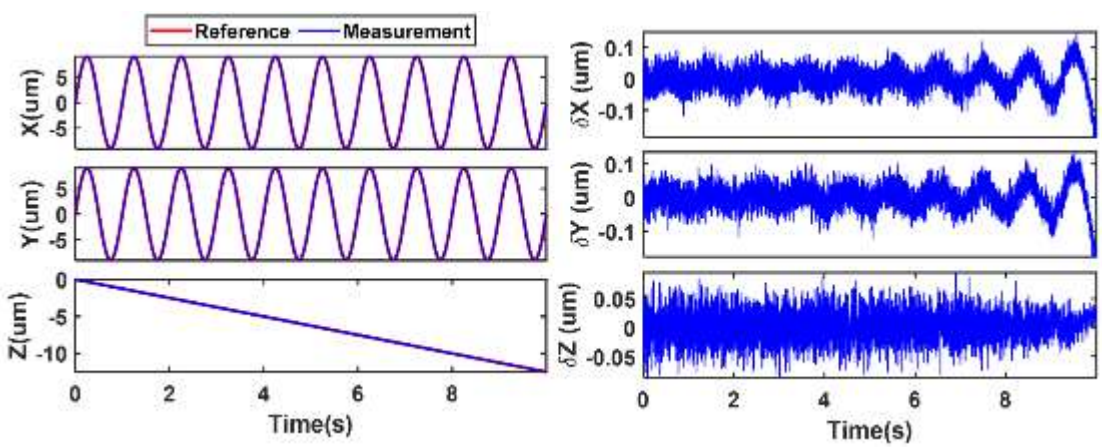

Fig. 30. The left panel shows the resulting motion using γ_N, and the right panel shows the corresponding tracking error

We repeated the experiment with real-time estimation and adaptive self-tuning control. Fig. 31 shows the result of motion control with real-time estimation and adaptive

self-tuning control. It is seen that the Brownian motion of the microbead is reduced when approaching the wall. The real-time estimated input gain is shown in the left panel of Fig. 32. It is compared to the real one. The real-time estimated input gain tracks closely with the real one. The real-time estimated input gain minimized the parameter error in (4.5). We separated the mean of the tracking error by using the IIR filter in (4.6) and the standard deviation of the random tracking error is calculated and compared with the theoretical value. The result is shown in the right panel of Fig. 32. The mean of the tracking error is zero for all three axes, and the standard deviation of the tracking error tracked closely to the theoretical value.

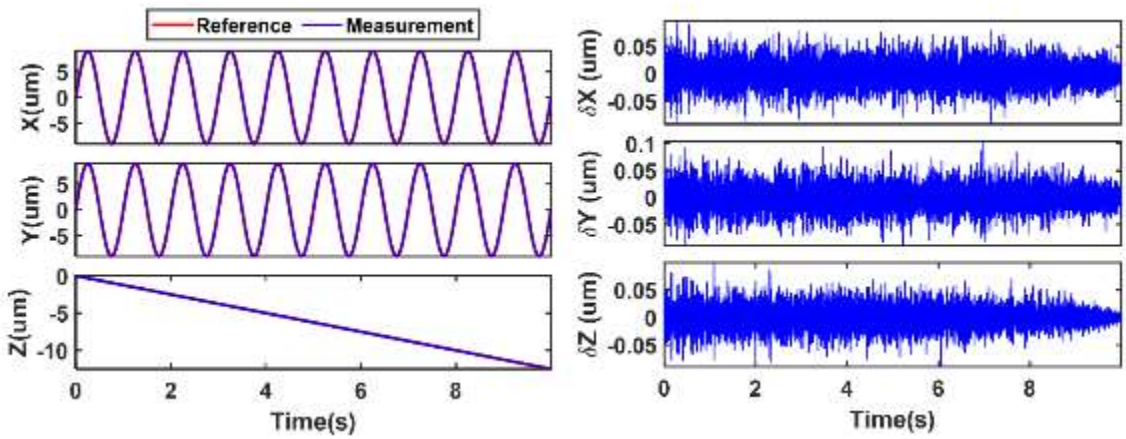

Fig. 31. The left panel shows the resulting motion using real-time estimated input gain, and the right panel shows the corresponding tracking error

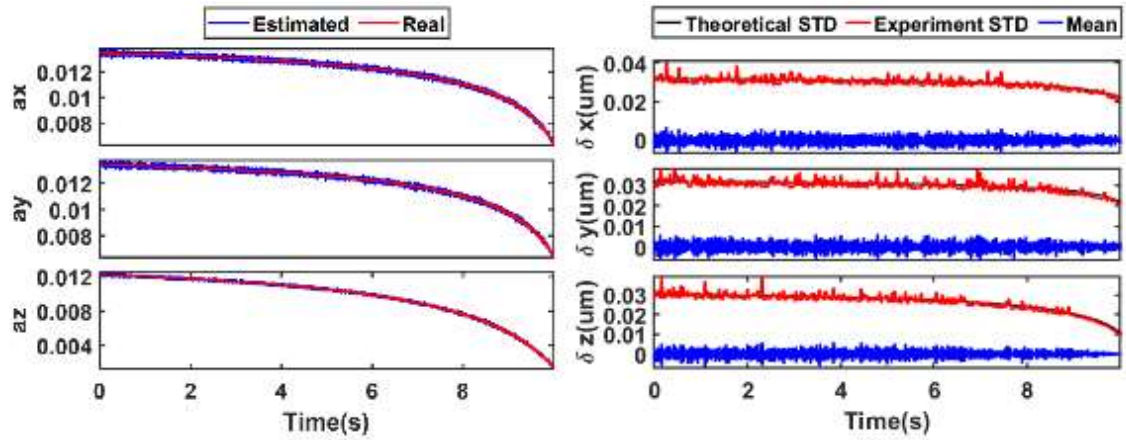

Fig. 32. The left panel shows the real-time estimation of the input gain, and the right panel shows the mean and the standard deviation of the tracking error.

From the previous experiment, it is important to implement real-time estimation and adaptive self-tuning control when the probe is near the wall. The next experiment will focus on the motion near the wall. The probe started from the initial position of $(0\ \mu m, 0\ \mu m, -11.5\ \mu m)$. The wall is placed at $x = -15$ micrometers. The z motion of the probe is a 3 Hz sinusoidal wave with 1 micrometer amplitude. Thus, the closest distance from the wall to the probe center is 2.5 micrometers. The motion of x and y are the same. They are 5 Hz sinusoidal waves with 9 micrometers amplitude. The total motion lasted for 3 seconds. The probe is stabilized and controlled by the control law (3.6) with nominal drag coefficient γ_N. The left panel of Fig. 33 shows the result of motion without using real-time estimation. The right panel of Fig. 33 shows the corresponding tracking error. The amplitude of the tracking error of x-axis and y-axis is around 0.8 micrometer. As the probe approached the wall, the tracking error of all three axes increased. The real drag coefficient at the top of the sinusoidal wave in the z-direction is

five times larger than the nominal drag coefficient. The drag coefficient increased significantly to twenty times larger than the nominal drag coefficient at the bottom of the sinusoidal wave, so the nominal drag coefficient γ_N in (3.6) is much smaller than the real drag coefficient $\gamma[k]$. The peak and valley of the tracking error happened at the bottom of the sinusoidal wave due to the largest parameter error.

We repeated the experiment with real-time estimation and adaptive self-tuning control. The left panel of Fig. 34 shows the result of motion control with real-time estimation and adaptive self-tuning control. The right panel of Fig. 34 shows the corresponding tracking error. The tracking error is significantly reduced as the real-time estimation and adaptive self-tuning control are applied. The Brownian motion is reduced when the probe approached the wall because the Brownian motion is inverse proportional to the drag coefficient. The real-time estimated input gain is shown in the left panel of Fig. 35. The real-time estimated input gain tracks closely with the real input gain. The real-time estimated input gain minimized the estimation error in (4.5). An IIR filter (4.6) is employed to separate the mean of the tracking error, and the standard deviation of the random tracking error is calculated and compared with the theoretical value. The result is shown in the right panel of Fig. 35. The mean of the tracking error is zero for all three axes, and the standard deviation of the tracking error tracked closely to the theoretical value.

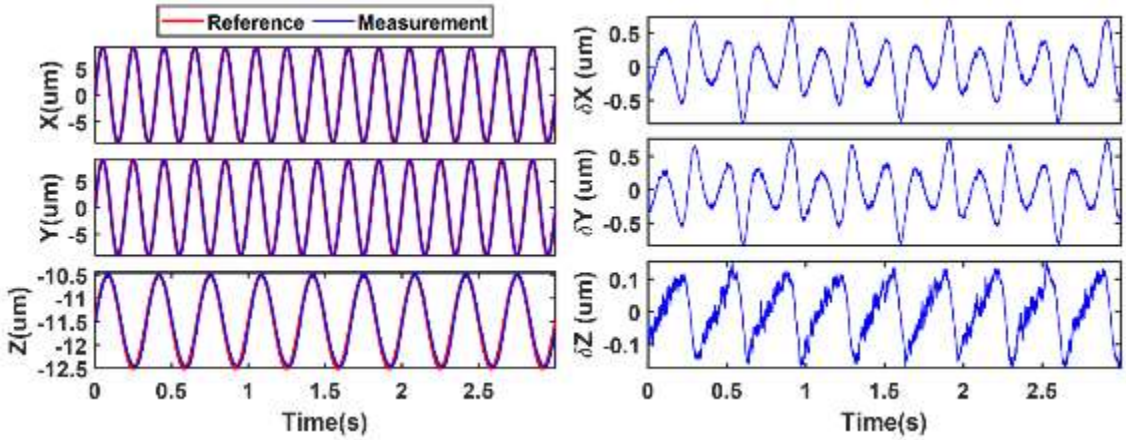

Fig. 33. The left panel shows the resulting motion using γ_N, and the right panel shows the corresponding tracking error

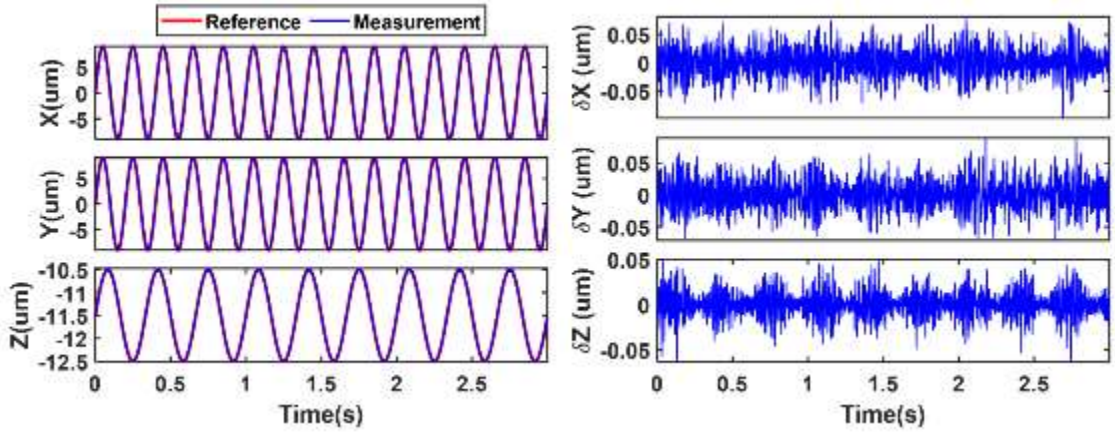

Fig. 34. The left panel shows the resulting motion using real-time estimated input gain, and the right panel shows the corresponding tracking error

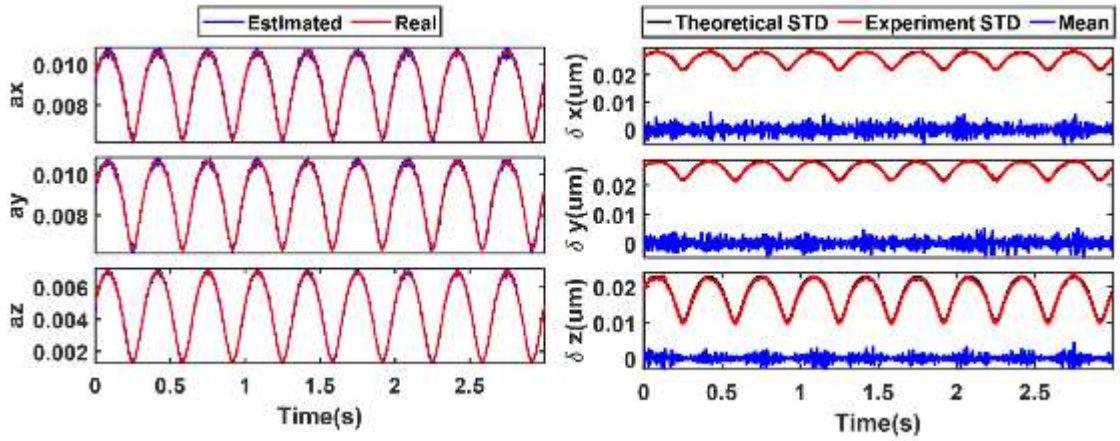

Fig. 35. The left panel shows the real-time estimation of the input gain, and the right panel shows the mean and the standard deviation of the tracking error.

The last experiment is three-dimensional rapid steering of the magnetic probe experiment in aqueous solutions utilizing real-time estimation and adaptive self-tuning control. The magnetic probe was steered to follow a designed conical helix in the workspace. The probe started from the initial position of $(0\ \mu m, 0\ \mu m, 0\ \mu m)$, was controlled to rotate along the upward pitch for five cycles, each cycle taking 0.35 seconds, reached the highest point $(30\ \mu m, 0\ \mu m, 15\ \mu m)$, and then was steered back to the initial position in 0.25 seconds. This is the same trajectory as shown in chapter 3.5.3. In chapter 3.5.3, we are updating the drag coefficient based on the theoretical wall effect model with known wall height. In this experiment, the real-time estimation of the input gain enabled adaptive self-tuning control. Fig.36 shows the experimental results of three-dimensional rapid steering when the real-time estimation of the input gain and adaptive self-tuning control were used to render high-speed visual servo control. The left panel presents the resulting 3-D motion, and the right panel presents the tracking errors of the three axes.

The left panel of Fig. 37 is the real-time estimation of the input gain. The real-time estimated input gain followed closely with the real input gain. We applied an IIR filter (4.6) to the tracking error to separate the mean of the tracking error and calculated the standard deviation of the random tracking error. As expected, the results show superior performance for rapid 3-D steering with no mean errors in any of the three axes. The standard deviation of the random error for each axis is close to the theoretical prediction.

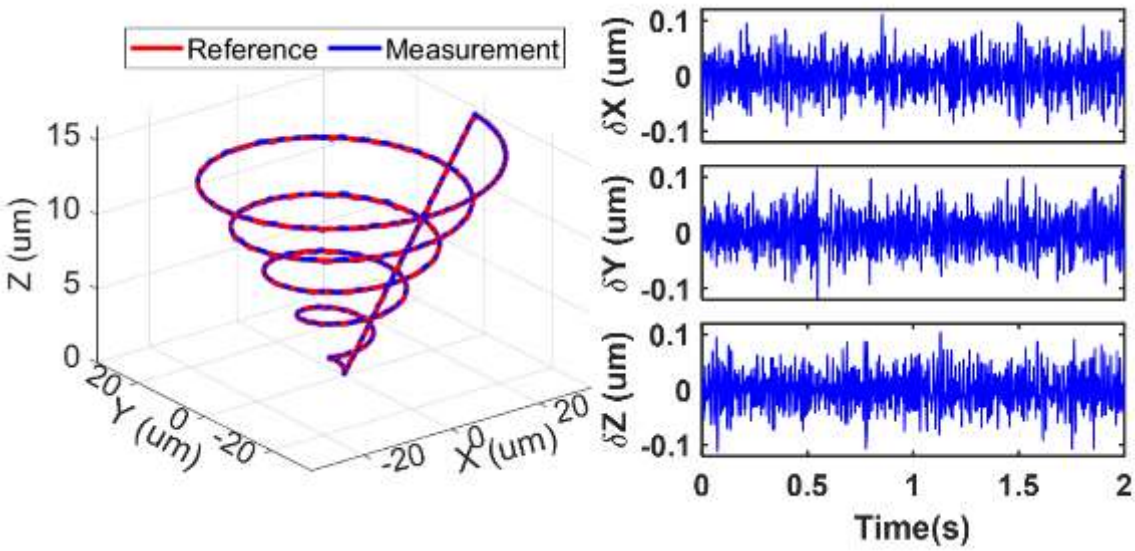

Fig. 36. Three-dimensional rapid steering: the left panel shows the resulting motion, and the right panel shows the tracking error.

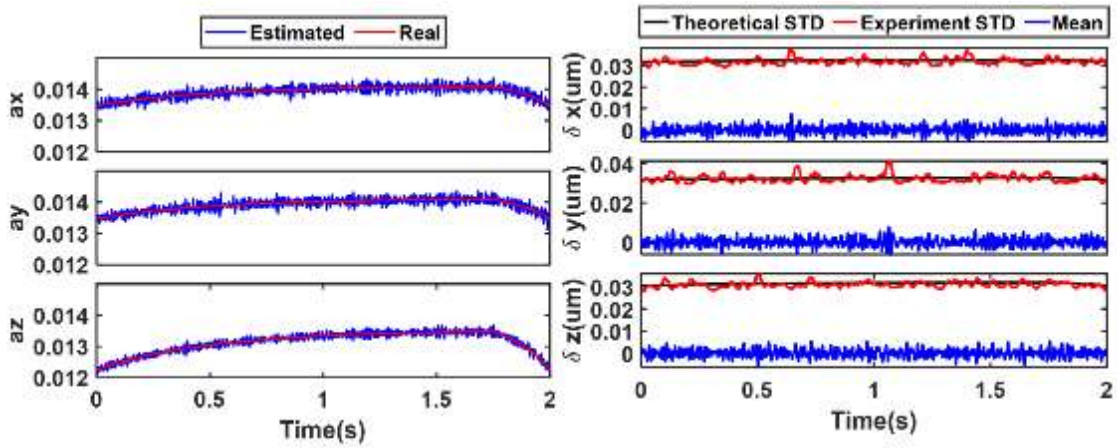

Fig. 37. The left panel shows the real-time estimation of the input gain, and the right panel shows the mean and the standard deviation of the tracking error.

4.4 Conclusion

The manipulation of the probe in aqueous solution is governed by the Langevin equation at low Reynolds numbers. The drag coefficient characterized by Stoke's law at low Reynold's number is not constant in aqueous solutions. As the magnetic microprobe approaches a wall, the drag coefficient increases with its distance from the surface. The position-dependent drag coefficient is characterized by two principal directions, vertical and parallel to the wall. Based on the model, a position-dependent control law was derived and implemented in chapter 3 to achieve high-speed steering of magnetic microprobe in aqueous solutions. We can achieve zero-mean tracking error and minimum variance control simultaneously. It is worth noting that the drag coefficient in the control law is updated by the theoretical value with known surface height.

However, the information of the surface height is not always available. If we use the nominal drag coefficient in the model-based control law, it might cause instability due

to the mismatch between the nominal drag coefficient and the real drag coefficient. The stability range is discussed and verified in chapter 4.2. We derived a Kalman filter-based augmented estimator that estimates the time-variant and/or spatially dependent input gain (drag coefficient) in the absence of surface height. It prevents instability and enables us to realize adaptive self-tuning control when propelling the probe in aqueous solution accounting for wall effects. Four experiments are conducted to validate the stability range and the tracking performance after applying the real-time input gain estimation and adaptive self-tuning control.

There are still several aspects that can be improved to enable the magnetic probe to steer in an unknown environment. In this chapter, we are discussing the scenario that the principal axis of the surface normal is overlapped with the z-direction of the measurement coordinate, so we can estimate the input gain of the three axes separately. This is not applicable for an unknown environment, i.e., incline surface, and spherical surface. The three separate input gains are extended to a 3x3 input matrix in the unknown environment. The estimation of the surface orientation and the estimation of the 3x3 input matrix is required to perform 3-D rapid steering with zero-mean tracking error and minimum variance control in unknown environments.